

View Reviews

Paper ID

318

Paper Title

Systematic Evaluation of Cross-Anatomy Transfer Learning in Fracture Detection: Performance Degradation, Clinical Implications, and Sample Efficiency

Track Name

AAI2026

Reviewer #1

Questions

1. {Summary} Please briefly summarize the main claims/contributions of the paper in your own words.

The paper presents the cross-anatomy with Transfer learning in medical imaging for 5 Anatomy including wrist (source), Elbow, Hand, Shoulder, and Finger. The DenseNet-121 has been pretrained on ImageNet as the backbone and adapted for binary fracture detection by replacing the final 1000-class layer with a single output class. neuron.

2. {Novelty} How novel are the concepts, problems addressed, or methods introduced in the paper?

Very Good

3. {Soundness} Is the paper technically sound?

Very Good

4. {Impact} How do you rate the likely impact of the paper in the research community?

Very Good

5. {Clarity} Is the paper well-organized and clearly written?

Yes

6. Detailed Feedback for the Authors.

The authors have presented Multi-seed validation, FN rate analysis, High-confidence errors, Asymmetric failures, and evaluated cross-anatomy transfer learning for fracture detection across five upper-extremity anatomies.

7. Overall evaluation

Strong Accept (As it is)

8. Recommendation (if accepted):

Full length Article (12+ pages)

Reviewer #2

Questions**1. {Summary} Please briefly summarize the main claims/contributions of the paper in your own words.**

The paper presents a systematic study on cross-anatomy transfer learning for fracture detection using medical imaging. It evaluates how models trained on one anatomical region perform on another and analyzes performance degradation, sample efficiency, and clinical implications. Experimental results compare transfer learning strategies and dataset sizes to determine optimal training approaches for medical fracture detection systems.

2. {Novelty} How novel are the concepts, problems addressed, or methods introduced in the paper?

Good

3. {Soundness} Is the paper technically sound?

Good

4. {Impact} How do you rate the likely impact of the paper in the research community?

Very Good

5. {Clarity} Is the paper well-organized and clearly written?

Yes

6. Detailed Feedback for the Authors.

The paper presents a systematic evaluation of cross-anatomy transfer learning for fracture detection using medical imaging datasets. The study is well motivated, as collecting labeled medical data for every anatomical region is difficult and expensive, and transfer learning can help reduce data requirements.

The experimental design is well structured, and the paper evaluates performance degradation when models trained on one anatomy are applied to another. The analysis of sample efficiency and training data size is a strong contribution. The results provide useful insights into when transfer learning is effective and when anatomy-specific training is required.

The paper is technically sound and the methodology is clearly explained. Figures, tables, and performance comparisons help in understanding the results. The clinical implications discussed in the paper add practical value to the research.

However, the paper could be improved by including more comparison with existing transfer learning approaches, additional datasets, and discussion on real-world deployment challenges. Some sections could also include more detailed explanation of model architecture and hyperparameters for reproducibility.

Overall, this is a strong and well-structured research paper with practical relevance in medical AI and fracture detection systems.

7. Overall evaluation

Accept (Minor Revision)

8. Recommendation (if accepted):

Full length Article (12+ pages)

Reviewer #4

Questions

1. {Summary} Please briefly summarize the main claims/contributions of the paper in your own words.

Main Claims & Contributions

The paper systematically evaluates zero-shot transfer learning for fracture detection across upper-extremity anatomies using the MURA dataset. Key contributions include:

Rigorous cross-anatomy evaluation — trains DenseNet-121 on wrist radiographs and evaluates on four target anatomies (elbow, hand, shoulder, finger) with 5-seed statistical validation and Bonferroni-corrected significance testing

Quantified performance degradation — demonstrates statistically significant

2. {Novelty} How novel are the concepts, problems addressed, or methods introduced in the paper?

Good

3. {Soundness} Is the paper technically sound?

Good

4. {Impact} How do you rate the likely impact of the paper in the research community?

Good

5. {Clarity} Is the paper well-organized and clearly written?

Yes

6. Detailed Feedback for the Authors.

Detailed Feedback for Authors

Strengths

Clinically motivated problem: Excellent framing of patient-safety risks from high-confidence misclassifications and asymmetric failures.

Rigorous methodology: 5-seed validation with Bonferroni correction, Cohen's d effect sizes, and 95% confidence intervals demonstrate statistical rigor.

Comprehensive clinical analysis: False negative rates, high-confidence error analysis, and operating-point characteristics provide actionable deployment guidance.

Clear visualizations: ROC, precision-recall curves, and learning curves effectively communicate findings.

Reproducibility: Detailed hyperparameters, dataset splits, and experimental protocol enable reproduction.

Critical Weaknesses

Limited architectural scope: Only DenseNet-121 evaluated. No comparison with ResNet-50, Vision Transformers, or modern architectures. Findings may not generalize to other models.

Single dataset limitation: MURA only. Cross-validation on external datasets (e.g., CheXpert, Stanford Radiology) would strengthen generalizability claims.

Shallow analysis of failure modes: Why does hand resist recovery? The paper speculates about bone morphology but provides no visualization (Grad-CAM, attention maps) or feature analysis to support claims.

No domain adaptation exploration: Paper mentions domain adaptation methods [5,14] but doesn't evaluate them. This limits practical utility — practitioners need solutions, not just problem characterization.

Sample efficiency plateau unexplained: Hand plateaus at 83.5% AUC with 500 samples. Is this a fundamental anatomical limit or a model capacity issue? Testing with larger models (ResNet-152, Vision Transformers) could clarify.

Missing confidence calibration analysis: The paper identifies high-confidence errors but doesn't evaluate calibration metrics (ECE, MCE) or propose calibration methods (temperature scaling, Platt scaling).

Limited operating-point guidance: While asymmetric failures are identified, the paper lacks c

7. Overall evaluation

Accept (Minor Revision)

8. Recommendation (if accepted):

Short Paper (8-12 pages)

Reviewer #5

Questions

1. {Summary} Please briefly summarize the main claims/contributions of the paper in your own words.

The paper systematically evaluates cross-anatomy transfer learning for fracture detection using deep learning on the MURA dataset, revealing significant performance

degradation (12–29% AUC drop) in zero-shot settings and high false negative rates, especially for hand and finger cases. It highlights clinical risks from overconfident errors and asymmetric failures, and quantifies sample efficiency, showing varied data needs (50–500+ samples) for reliable adaptation.

2. {Novelty} How novel are the concepts, problems addressed, or methods introduced in the paper?

Good

3. {Soundness} Is the paper technically sound?

Good

4. {Impact} How do you rate the likely impact of the paper in the research community?

Very Good

5. {Clarity} Is the paper well-organized and clearly written?

Yes

6. Detailed Feedback for the Authors.

The manuscript presents a rigorous and clinically relevant evaluation of cross-anatomy transfer learning for fracture detection, offering significant contributions through systematic experimentation, multi-seed validation, and detailed clinical impact analysis. The use of the MURA dataset and DenseNet-121 is appropriate, and the findings—particularly the substantial AUC degradation, high false negative rates, and prevalence of high-confidence errors—are both insightful and impactful for real-world deployment. The inclusion of sample efficiency analysis further strengthens the practical relevance of the study by quantifying data requirements across anatomies. However, the manuscript could be improved in several areas. First, while the experimental design is strong, the rationale for selecting wrist as the sole source anatomy could be better justified. Second, the methodology section would benefit from additional clarity regarding preprocessing choices and the absence of data augmentation, as this may affect generalizability. Third, although the results are comprehensive, the discussion could be deepened by linking findings more explicitly to existing domain adaptation literature and potential mitigation strategies such as calibration or uncertainty estimation. Additionally, visualizations (e.g., ROC and PR curves shown in figures on pages 6–8) are informative but would benefit from more interpretive explanation to guide readers. The limitations are acknowledged, but expanding on external validation and alternative model architectures (e.g., transformers) would strengthen the study. Finally, the conclusion effectively summarizes contributions, yet it could more clearly articulate actionable guidelines for clinicians and AI practitioners. Overall, the paper is a valuable contribution with strong empirical grounding but would benefit from enhanced clarity, deeper discussion, and broader contextualization.

7. Overall evaluation

Accept (Minor Revision)

8. Recommendation (if accepted):

Full length Article (12+ pages)

